

Parcial No. 1

Programación Paralela

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Los códigos los puse aquí https://github.com/osvarp/paralelComputing_week3

Bajo la carpeta de 1midterm

¿Cómo influye la programación paralela, utilizando OpenMP ó Pthreads y MPI (En un entorno simulado con docker), en el tiempo de ejecución del filtrado de imágenes PPM y PGM en comparación con una implementación secuencial? **Lo hace mas rápido (a ratos).**

Versión secuencial

```
japeto@24b9df4f242e:~/app$ g++ main.cpp kernel.cpp lineal/map_pgm.cpp lineal/map_ppm.cpp -o lin.out
japeto@24b9df4f242e:~/app$ time ./lin.out images/lena.pgm lena_blur.pgm

real    0m0.245s
user    0m0.087s
sys     0m0.023s
japeto@24b9df4f242e:~/app$ time ./lin.out images/fruit.pgm fruit_blur.pgm

real    0m0.584s
user    0m0.192s
sys     0m0.023s
japeto@24b9df4f242e:~/app$ time ./lin.out images/puj.pgm puj_blur.pgm

real    0m0.834s
user    0m0.374s
sys     0m0.041s
japeto@24b9df4f242e:~/app$ time ./lin.out images/damma.pgm damma_blur.pgm

real    0m1.200s
user    0m0.450s
sys     0m0.064s
japeto@24b9df4f242e:~/app$ time ./lin.out images/sulfur.pgm sulfur_blur.pgm

real    0m0.728s
user    0m0.309s
sys     0m0.012s
```

Versión pthreads

```
japeto@24b9df4f242e:~/app$ g++ -pthread main.cpp kernel.cpp pthread/map_pgm.cpp pthread/map_ppm.cpp -o pthr.out
japeto@24b9df4f242e:~/app$ time ./pthr.out images/lena.pgm lena_blur.pgm

real    0m0.267s
user    0m0.134s
sys     0m0.032s
japeto@24b9df4f242e:~/app$ time ./pthr.out images/fruit.pgm fruit_blur.pgm

real    0m0.323s
user    0m0.159s
sys     0m0.021s
japeto@24b9df4f242e:~/app$ time ./pthr.out images/puj.pgm puj_blur.pgm

real    0m0.955s
user    0m0.644s
sys     0m0.020s
japeto@24b9df4f242e:~/app$ time ./pthr.out images/damma.pgm damma_blur.pgm

real    0m0.923s
user    0m0.737s
sys     0m0.019s
japeto@24b9df4f242e:~/app$ time ./pthr.out images/sulfur.pgm sulfur_blur.pgm

real    0m0.666s
user    0m0.470s
sys     0m0.028s
```

Versión OpenMP

```
japeto@24b9df4f242e:~/app$ g++ -fopenmp main.cpp kernel.cpp openMp/map_pgm.cpp openMp/map_ppm.cpp -o omp.out
japeto@24b9df4f242e:~/app$ time ./omp.out images/lena.pgm lena_blur.pgm

real    0m0.311s
user    0m0.977s
sys     0m0.020s
japeto@24b9df4f242e:~/app$ time ./omp.out images/fruit.pgm fruit_blur.pgm

real    0m0.462s
user    0m1.333s
sys     0m0.041s
japeto@24b9df4f242e:~/app$ time ./omp.out images/puj.pgm puj_blur.pgm

real    0m1.035s
user    0m3.157s
sys     0m0.050s
japeto@24b9df4f242e:~/app$ time ./omp.out images/damma.pgm damma_blur.pgm

real    0m1.409s
user    0m3.984s
sys     0m0.040s
japeto@24b9df4f242e:~/app$ time ./omp.out images/sulfur.pgm sulfur_blur.pgm

real    0m0.701s
user    0m2.261s
sys     0m0.040s
```